

Research Briefing | Euro Area and UK

How much are indirect effects of energy still lifting core inflation?

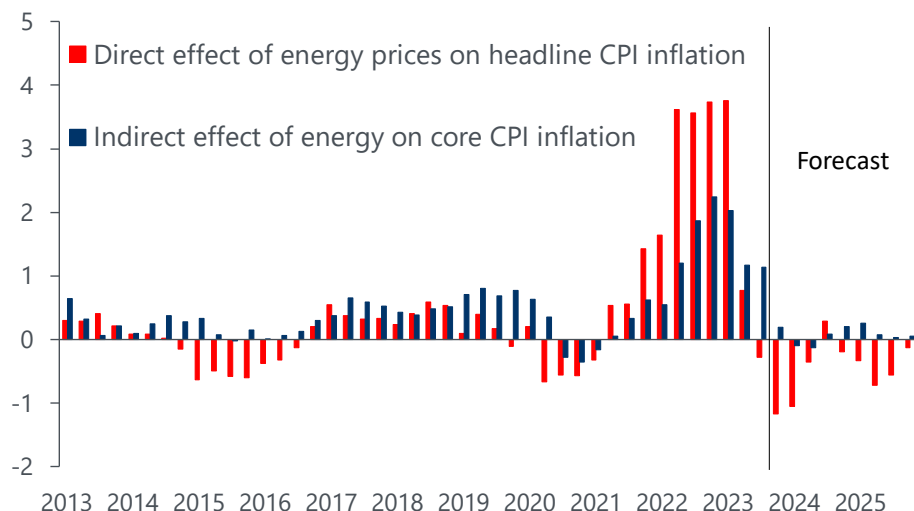
- The indirect effects of energy prices on core inflation have diminished since late 2022/early 2023, but remain sizeable, and are currently adding about 1pp to core inflation in the UK and the euro area.
- Direct effects of energy on headline CPI inflation have turned negative in the UK and the euro area and, even with the recent rise in oil prices, are likely to be more negative in coming months. The indirect effects on inflation from energy prices tend to lag slightly but -- barring a new surge in energy costs -- are likely to shrink in the next few months and probably turn negative in early 2024 in the UK and the euro area.
- These effects from energy prices, together with easing global supply constraints for goods and disinflationary effects as tight monetary policy dampens activity, are likely to bring headline and core inflation markedly lower in the next few quarters in the euro area and the UK.
- With weak economic growth and falling inflation, policy rates have probably peaked in the UK and the euro area. In judging whether and when to ease, the European Central Bank and the Bank of England are likely to focus on whether inflation is heading back to target on a sustained basis, rather than just consider the latest headline rate. Trends in pay growth and core inflation excluding indirect energy effects are likely to be useful gauges to assess when these conditions for a sustained return of inflation to target are in place.

At the start of this year, we argued that the [indirect effects on core and headline inflation](#) of the energy price surge in the UK and the euro area would remain large in the near term and would probably keep core inflation elevated during the first two to three quarters of this year, but fade thereafter. This piece updates that work. Our key message is that outlook remains broadly on track. Indirect effects from energy prices have stayed quite high in the last couple of quarters but (barring a new surge in energy prices) these indirect effects are likely to decline markedly in the next quarter or two, contributing to a marked drop in headline and core inflation in the UK and the euro area.

Chart 1: Indirect effects of energy prices on core inflation likely to fall sharply in next quarter or two

UK: Direct and Indirect Effects of Energy on CPI

% ppts



Sources: Oxford Economics / Haver Analytics

Indirect effects of energy prices

Over the last couple of years, the direct effects of higher energy prices – through increases in households' prices for petrol, electricity, and gas – have added substantially to CPI inflation in the UK and the euro area. At the peak in 2022/ 2023, these direct effects added 4.1pps to inflation in the UK and 4.8pps in the euro area. With energy prices lower in recent quarters, these direct effects on CPI inflation are now turning negative, and in the latest data from August were -0.3pps in the UK and the euro area.

The energy price surge also had indirect effects on core inflation – excluding food, alcohol, tobacco, and energy – because of the impact on firms' costs in a wide range of sectors. For example, higher energy prices meant that restaurants faced higher costs for heating, lighting, and cooking, while their food costs rose markedly, partly because food production is relatively energy-intensive. As a result, higher energy prices have caused a sharp rise in catering prices in the CPI, a part of core inflation and services inflation.

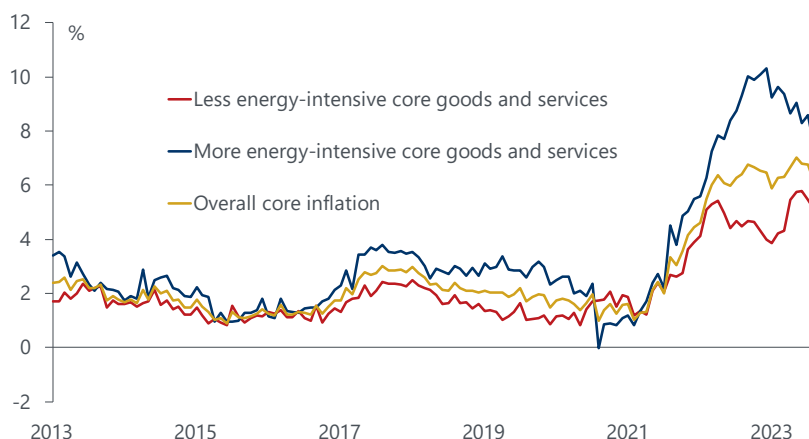
[Early this year](#), we showed how to track these indirect effects. Using input-output tables for the UK, we split core CPI items into those for which energy is a relatively high share of total costs, including the share of energy in firms' purchases of goods and services, and those for which energy is a relatively low share of total costs. For example, among services items, hotels and restaurants are relatively energy-intensive, while financial services, home insurance, and hairdressing have relatively low energy intensity. A proxy for the indirect effects of energy prices on core inflation is the gap between overall core inflation and the inflation of core items with relatively low energy intensity – this equals the contribution to core inflation of "excess" inflation in items with high energy intensity.

For the euro area, we use a similar approach, based on the split of CPI items by energy intensity [discussed late last year](#) by ECB Chief Economist Phillip Lane. For the euro area, the items with relatively high energy-intensity account for about 40% of the overall core CPI.

As **Charts 2 and 3** show, in the UK and the euro area, the energy price surge was followed by a marked jump in the weighted average inflation rate of goods and services with relatively high energy intensity, with a much smaller rise in inflation for items of relatively low energy intensity.

Chart 2: Inflation in energy-intensive core CPI items rose sharply last year, but is now slowing

UK: Split of core CPI inflation by energy intensity

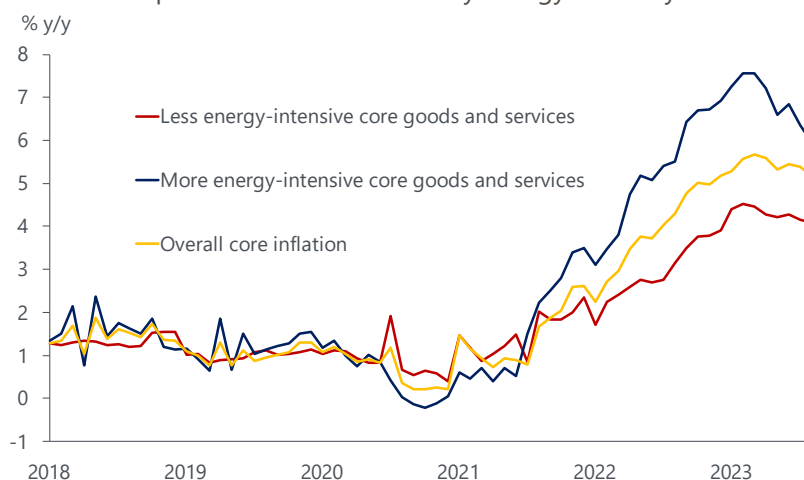


Sources: Oxford Economics/Haver Analytics

Indirect effects of energy prices

Chart 3: Inflation in energy-intensive items slowing in the euro area

Euro Area: Split of core CPI inflation by energy intensity



Sources: Oxford Economics/Haver Analytics

Based on this, we estimate that the indirect effects added about 2pps to core inflation in the UK and 1-2pps to core inflation in the euro area at the peak in late 2022/early 2023. The effects on headline inflation were similar, allowing for the indirect effects of energy prices on noncore items such as food and drink. These indirect effects have been quite large for both core goods and services, but slightly greater for goods than services, on average. So far, these indirect effects are roughly in line with what you would expect given the scale of the energy price shock. On average, indirect effects from energy prices on core inflation tend to be about half as big as the direct effects on CPI inflation, but slightly lag, typically coming through over two to three quarters after the direct effects.

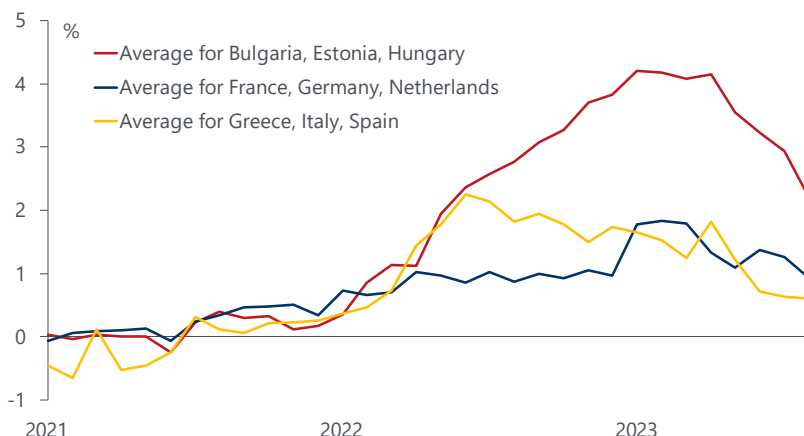
If anything, our estimates may understate the indirect effects of energy prices because our calculations implicitly assume that energy costs have accounted for none of the rise in inflation for items with low energy intensity, whereas in practice, it is likely that they did add something. [A paper by Banca D'Italia](#) highlights the sizeable indirect effects of energy on core inflation, using various econometric techniques.

The scale of the indirect effects of energy prices on core inflation in 2022-2023 vary across EU countries, partly reflecting differences in the scale of increases in domestic energy prices. For example, between H1 2021 and H2 2022, typical gas prices for business users rose by more than 400% on average in Bulgaria, Greece, and Estonia. In these countries, indirect effects of energy on core inflation were relatively large, exceeding 4pps over the winter of 2022/2023. In Greece, Italy, and Spain, business gas prices rose by an average of about 300% over the same period, and indirect effects on core inflation exceeded 2pps last year. Conversely, in Germany, France, and the Netherlands, business gas prices rose by an average of 80% and the indirect effects of energy on inflation have generally been smaller, apart from a blip over last winter.

Indirect effects of energy prices

Chart 4: Indirect effects of energy prices on core inflation vary across EU countries

EU: indirect effects of energy prices on core inflation in selected countries



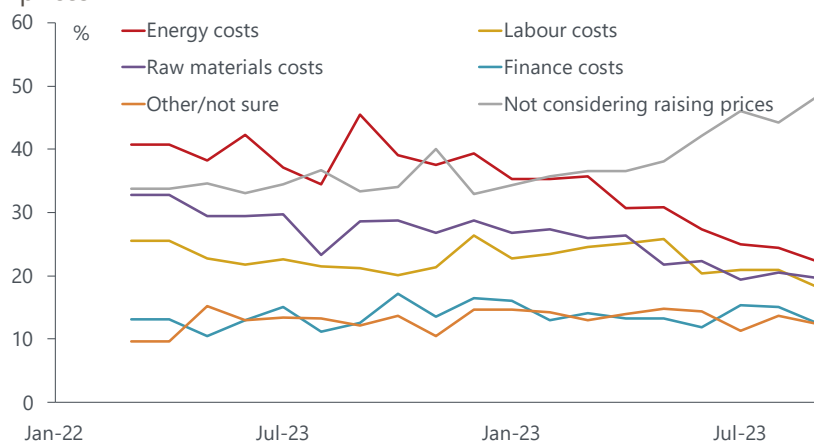
Sources: Oxford Economics/Haver Analytics

These indirect effects are still quite large, adding about 1pp to y/y core and headline inflation in the UK and the euro area in August, the latest data. But they are smaller than early this year, with a notable slowdown in inflation for items which are relatively energy intensive in the UK and the euro area. This has played a major role in the slight reduction in core inflation in recent months. Together with the reduced direct effect from energy prices, we estimate that the q/q annualised inflation rate, seasonally adjusted, has fallen to about 3% in the UK and the euro area.

Results from the Office for National Statistics' Business Insights and Conditions Survey highlight the wide-ranging effects from energy costs lifting UK firms' prices over the last 18 months, and the extent to which these pressures are easing. On average in 2022, energy costs were the top factor causing firms to lift selling prices in almost every sector of the economy – manufacturing and construction were exceptions, with raw material prices as the top factor. However, the share of firms that report energy costs as a key factor that caused them to lift selling prices has fallen to 22% from 46% in Q3 2022 and now carries a similar weight to pay growth. With less impetus from energy costs, a growing share of firms report they do not intend to lift their selling prices, signalling lower inflation.

Chart 5: Higher energy prices were the top factor causing firms to raise selling prices last year

UK: Pct of firms citing various factors as reason to raise selling prices



Sources: Oxford Economics/Haver Analytics

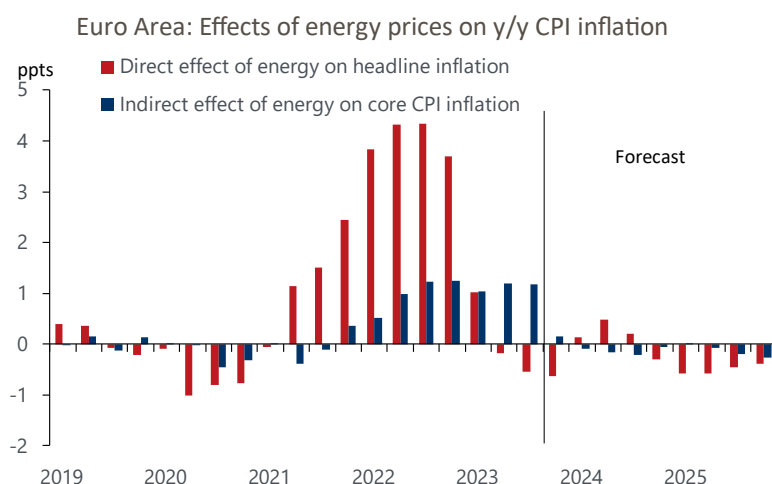
Given favourable base effects, the drop in wholesale gas prices since late 2022, and assuming wholesale energy costs track forward prices, the direct effects of energy on CPI will probably turn more substantially negative, cutting close to 1pp off euro area inflation in Q4 and more than 1pp off UK CPI inflation in Q4 this

Indirect effects of energy prices

year and Q1 2024. These estimates take into account the recent rise in oil prices, which has a modest effect compared to last year's energy price shock, but obviously would need to be revised if energy prices surge again.

Given the usual lags, as the recent decline in energy inflation and the decline that is likely in coming months feed through to firms' costs (and barring a new surge in energy prices), indirect effects from energy prices on core and headline inflation are set to diminish sharply, falling to roughly zero in the UK and the euro area in Q4, and slightly negative in early 2024.

Chart 6: Reduced energy effect will bring core inflation markedly lower in the next few quarters



Sources: Oxford Economics / Haver Analytics

The main implications

- While it may be tempting to regard core inflation and services inflation as "clean" measures of inflation that are not affected by energy prices and other global shocks, in practice, this is not correct. The indirect effects of energy prices on core inflation are smaller than the direct effects of energy on headline inflation, but they are still sizeable and have added substantially to core inflation in recent months, and this applies to services as well as goods.
- With the prospect of a reduction of the direct and indirect effects of energy prices, plus additional disinflationary pressure on goods prices as global supply constraints ease, and some initial impact from weak economic growth and rising spare capacity, y/y inflation in headline and core is likely to fall markedly in the UK and the euro area over the next few quarters. We expect headline inflation at the end of this year to be down to about 3% in the euro area and about 4.5% in the UK. Barring a new surge in energy prices, we forecast inflation will return to the 2% targets around mid-2024 for the euro area and late-2024 for the UK. We stress that only a modest part of this expected slowdown reflects our expectations for pay growth and spare capacity, with a much bigger role from energy prices – direct and indirect effects – and the global slowdown in manufactured goods prices.
- With the prospect of lower headline and core inflation, as well as clear signs of weakness in business surveys, we believe that the ECB and the BoE tightening cycles have ended, and expect both central banks to keep policy rates stable, at restrictive levels, over the winter.
- Both central banks may face a difficult balancing act over the next 12-18 months in judging whether, when, and how much to cut rates. In the UK and the euro area, the coming few quarters will likely see little or no economic growth with possible recessions, markedly lower headline inflation, and some decline in core inflation, which may help reduce inflation expectations. At face value, such conditions could point to considerable scope for lower rates during 2024 and a relatively prompt return to a more neutral stance. However, in practice, the response of the ECB and the BoE will not just depend on how quickly inflation is approaching 2%, it will depend mainly on their assessment of inflation prospects – especially whether they judge lower headline and core inflation to be persistent. Easing is less likely if

Indirect effects of energy prices

the drop in inflation mainly reflects direct and indirect effects from energy because these disinflationary effects will eventually fade.

Going forward

In judging the monetary policy outlook, it probably will be useful to focus on pay growth and core inflation measures that strip out indirect effects of energy prices, such as "super-core" measures. Currently, these gauges remain well above a target-consistent pace in the UK and the euro area. The ECB and the BoE will probably need to see significant slowdowns in such measures, as well as headline inflation, to consider cutting interest rates.

Our central forecast is that the ECB will start to cut rates in Q2 next year, with the BoE following about a quarter later, but there are risks on both sides of this forecast. Easing could come earlier if these central banks become more confident that lower inflation will persist, for example if pay growth and inflation of less energy-intensive items head markedly lower. Conversely, if pay growth and such super-core inflation gauges remain relatively high, as now, then the ECB and BoE will probably be reluctant to cut rates much, if at all, unless activity data worsen dramatically.